EE445 Mod2-Lec3: Least Squares Regression

References:

- [VMLS]: Chapter 12

Agenda

- Optimality via linear algebra (eigenvalues of Hessian & linear independence)
- Least squares solution via normal equations

Linear Models: Predicting Outputs from Inputs

- **Goal**: learn a function (aka "model") $f_\theta(x)$ that predicts quantity $y \in \mathbb{R}$ from input $x \in \mathbb{R}^d$.
- For linear models:

$$f_\theta(x) = \theta^\top x = \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d \qquad \text{(linear)}$$

$$f_\theta(x) = \theta^\top \begin{bmatrix} x \\ 1 \end{bmatrix} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d \qquad \text{(affine)}$$

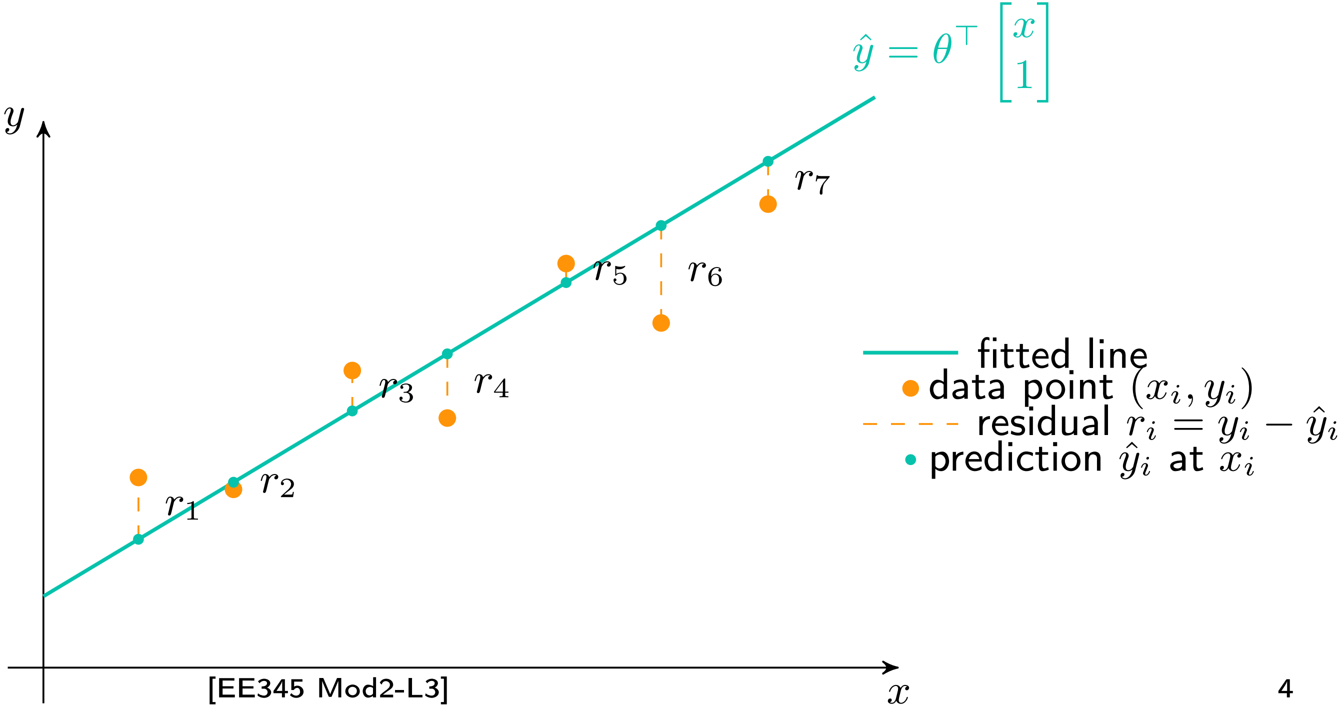
- Given data (x_i, y_i) for $i = 1, \ldots, n$, we choose θ to minimize prediction error.
- Examples:
  - Predict house price from size, bedrooms, age.
  - Predict exam score from study hours.

Measuring Predictor Error in 2D (Linear Regression)

For a fitted line $\hat{y} = \theta_1 x + \theta_0$ and points (x_i, y_i), the **residual** is

$$r_i \;=\; y_i - \hat{y}_i \;=\; y_i - \theta x_i - \theta_0, \quad \text{and} \quad \mathcal{L}(\theta_1, \theta_0) = \sum_{i=1}^{n} r_i^2.$$

OLS loss: minimize total squared residual length $\sum_i \|r_i\|_2^2$ (vertical errors).



Empirical Risk Minimization (ERM): Quientissential ML Problem

Goal of supervised learning: Find a function that makes good predictions on new data.

Challenge: We don't know the true data distribution; indeed, we only see a finite set of samples.

Idea of ERM:

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \ell(f_\theta(x_i), y_i) \quad \textbf{average over } n \textbf{ samples} = \textbf{empirical risk}$$

Least Squares (Linear Regression) Set-up

- Consider an $n \times d$ matrix $X \in \mathbb{R}^{n \times d}$ and vectors $y \in \mathbb{R}^n$, $\theta \in \mathbb{R}^d$

Least Squares Optimization Problem

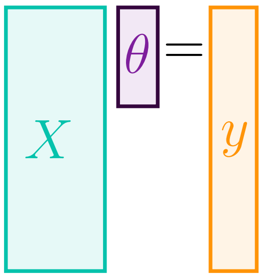
Find θ such that $X\theta = y$ that minimizes the ℓ_2 loss:

$$\min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta) = \min_{\theta \in \mathbb{R}^d} \|y - X\theta\|_2^2$$

- X is a matrix of training data; e.g., n is number of samples, d is number of 'features'
- $y \in \mathbb{R}^n$ contains 'target values', 'labels', or observations of real world phenomena
- $\theta \in \mathbb{R}^d$ is a set of feature weights

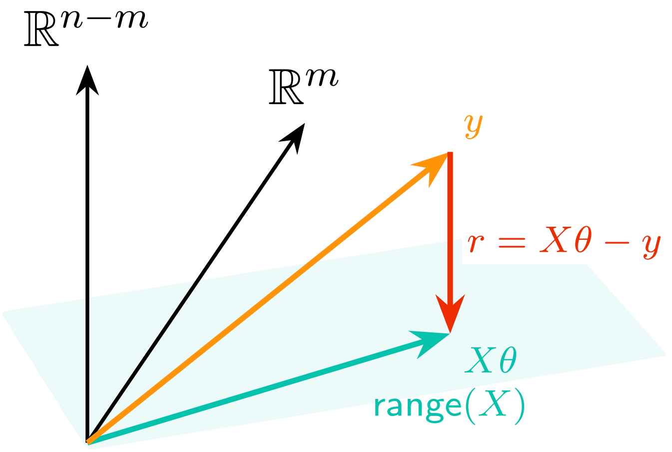
Overdetermined System of Equations $n \gg d$

- **Goal**: Find a solution to $X\theta = y$—that is, find θ such that $X\theta = y$
- **Overdetermined System**: X is a 'tall' matrix $\implies$ there are more samples (n) than features to choose (d).



$$\underbrace{\begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix}}_{\text{col } 1} \theta_1 + \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} \theta_2 + \cdots + \begin{bmatrix} x_{1d} \\ x_{2d} \\ \vdots \\ x_{nd} \end{bmatrix} \theta_d = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Projection Geometry: $y = X\theta + r$ with $r \perp \mathrm{range}(X)$



$m = \dim(\mathrm{range}(X))$

Finding Minima and Convexity

Goal: Find $\theta^\star$ that minimizes a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$.

Conditions for Minima [VMLS, App. C]

$$\text{first order condition:} \quad \nabla f(\theta^\star) = 0 \qquad \text{(stationary point)}$$

$$\text{Second-order condition :} \quad v^\top \underbrace{\nabla^2 f(\theta^\star)}_{=:H(\theta^\star)} v > 0 \quad \text{for all } v \neq 0, \qquad \text{(convexity)}$$

- Gradient $=$ direction of steepest ascent.
- Setting it to zero $\implies$ flat point.
- Positive-definite Hessian $\implies$ surface curves upward in all directions.

- $H(\theta) = \nabla^2 f(\theta)$ is the Hessian matrix.
- we say f is **strictly convex** near $\theta^\star$
- $\theta^\star$ is the unique minimum.

Example: Minima in Two Dimensions

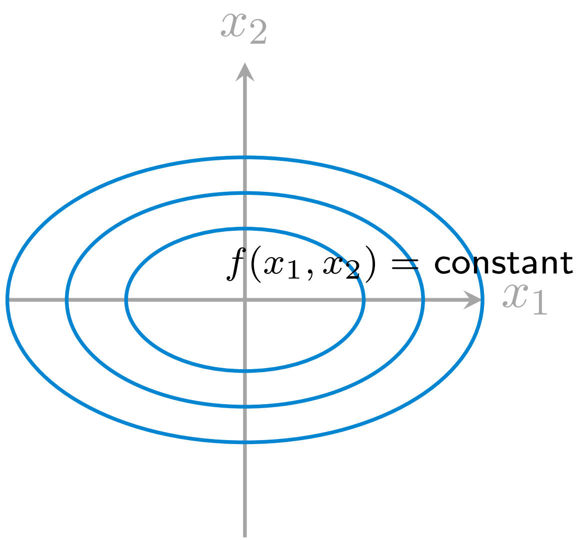
Now $f : \mathbb{R}^2 \to \mathbb{R}$ has a gradient and Hessian:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}, \quad H = \nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}.$$

Example: $f(x_1, x_2) = ax_1^2 + bx_2^2$ for $a, b > 0$.

$$H = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \implies v^\top H v = av_1^2 + bv_2^2.$$

$\implies H \succ 0$ (i.e. positive definite) if $a > 0$ and $b > 0$.



A More General 2D Hessian: Tilted Ellipses

Example (coupled quadratic):

$$f(x_1, x_2) = \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \underbrace{\begin{bmatrix} a & c \\ c & b \end{bmatrix}}_{H} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad (a, b > 0).$$

Gradient & Hessian:

$$\nabla f(x) = H\, x, \qquad \nabla^2 f(x) = H = \begin{bmatrix} a & c \\ c & b \end{bmatrix}.$$

General Convexity Condition via Eigenvalues

For any symmetric Hessian $H \in \mathbb{R}^{d \times d}$,

$$H \text{ is positive definite} \iff v^\top H v > 0 \text{ for all } v \neq 0.$$

Checking this directly is hard, so we use **eigenvalues**.

General Convexity Condition via Eigenvalues

Diagonalizing Symmetric Matrices (Spectral Theorem)

For general symmetric matrix H, there exists Q such that $QQ^\top = I$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ such that $H = Q\Lambda Q^\top$.

What we will show:

Linear Algebra Concepts

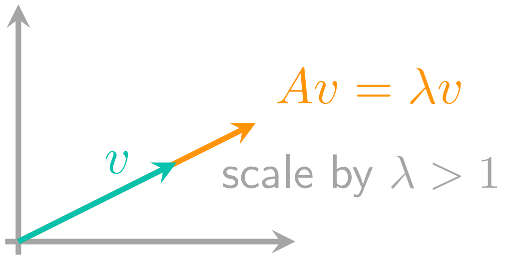
- Eigenvalues and eigenvectors
- Linear independence of eigenvectors
- Basis of eigenvectors
- Diagonalizing symmetric matrices
- Spectral theorem
- Positive definiteness

Eigenvalues and Eigenvectors

Definition of Eigenvalues and Eigenvectors

Geometric meaning:

- A scales (stretches or flips) v without changing its direction.
- The eigenvalues describe how much A stretches space along special directions.
- If $\lambda = 0$, that direction is *flattened* $\rightarrow$ indicates "rank deficiency".



Independence of Eigenvectors and Basis Formation

Linear Independence

Examples: Checking Linear Independence

Example 1

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Check if $\{v_1, v_2, v_3\}$ are linearly independent in $\mathbb{R}^2$.

Examples: Checking Linear Independence

Example 2

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Independence of Eigenvectors and Basis Formation

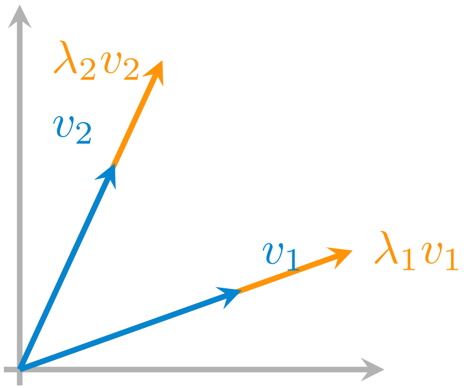
Theorem

If $Av_i = \lambda_i v_i$ and $Av_j = \lambda_j v_j$ with $\lambda_i \neq \lambda_j$, then v_i and v_j are linearly independent.

Independence of Eigenvectors and Basis Formation

Facts

- For $A \in \mathbb{R}^{d \times d}$ with d distinct eigenvalues, its eigenvectors $v_1, \ldots, v_d$ are linearly independent.
- If A has d distinct eigenvalues, its eigenvectors form a **basis** for $\mathbb{R}^d$.
- Distinct eigenvalues guarantee independent eigenvectors, which form a basis that diagonalizes A.



What is a Basis?

Definition of Basis

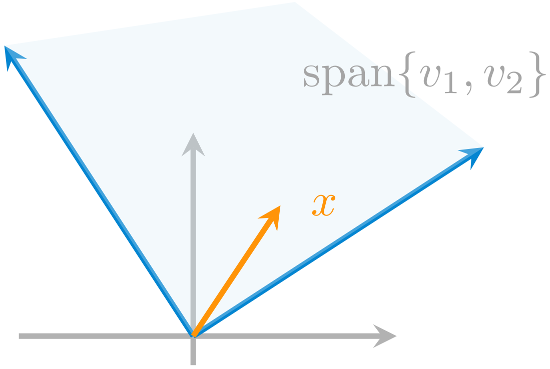
A *basis* of a vector space $V \subseteq \mathbb{R}^d$ is a set of vectors $\{v_1, v_2, \ldots, v_d\}$ such that:

(1) $v_1, \ldots, v_d$ are linearly independent, (2) they span V.

(2) $\iff$ every vector $x \in V$ can be written *uniquely* as $x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_d v_d$.

Key Ideas:

- "Span" $\implies$ the basis vectors *reach* all of V.
- "Linear independence" $\implies$ no vector in the set is redundant.
- The number of basis vectors = **dimension** of V.



Independence of Eigenvectors and Basis Formation

Why this matters:

- If $A \in \mathbb{R}^{d \times d}$ has d linearly independent eigenvectors, they form a **basis** for $\mathbb{R}^d$.
- In that basis, A acts simply by scaling each basis vector:

$$Av_i = \lambda_i v_i.$$

Diagonalization

Where are we

$A = V \Lambda V^{-1} \leftarrow \qquad H = \underline{Q \Lambda Q^T} \qquad \underline{Q^T Q = I}$

V is contains eigenvector of A

$\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$

- We have defined linear independence, and even shown how to diagonalize a matrix if it has independent eigenvectors.
- Now we want to come back to this Hessian matrix and questions about curvature.
- To apply the eigenvalue check, we need to show that Hessians can be decomposed as $H = A^\top A$ for some A.

Why Symmetric Matrices Can Be Diagonalized

$\langle v, w \rangle \qquad \langle v, \alpha w \rangle = \alpha^* \langle v, w \rangle$

Claim (Spectral Theorem)

If $H \in \mathbb{R}^{d\times d}$ is **symmetric** ($H = H^\top$), then there exists an **orthogonal matrix** Q (i.e., $Q^\top Q = I$) and a diagonal matrix Λ such that

$Q = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_d \\ | & & | \end{bmatrix}$

$$H = Q\Lambda Q^\top, \quad \text{where } \Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d), \text{ and } Hq_i = \lambda_i q_i.$$

- Symmetric matrices have real-valued eigenvalues
- Symmetry $\Rightarrow$ all eigenvectors corresponding to distinct eigenvalues ($\lambda_i \neq \lambda_j$) are orthogonal ($\langle v_i, v_j \rangle = v_i^\top v_j = 0$)
- Concluding eigenvectors form orthogonal basis for $\mathbb{R}^d$
  - conclude $Q^\top Q = I \Rightarrow$ [$HQ = Q\Lambda \Rightarrow H = Q\Lambda Q^\top$]

Why Eigenvectors of a Symmetric Matrix Are Orthogonal?

Let $H = H^T$; suppose

$H v_i = \lambda_i v_i \qquad H v_j = \lambda_j v_j \qquad$ s.t. $\lambda_i \neq \lambda_j$

LHS = Left hand side
RHS = right hand side

$\langle Hx, y\rangle = \langle x, Hy\rangle \quad \left[\text{LHS} = (Hx)^T y = x^T H^T y, \ \text{RHS} = x^T (Hy) = x^T H y \right.$

$$\begin{aligned}
\langle H v_i, v_j\rangle &= \langle \lambda_i v_i, v_j\rangle = \lambda_i v_i^T v_j \\
\parallel \qquad & \qquad\qquad\qquad\qquad \parallel \\
\langle v_i, H v_j\rangle &= \langle v_i, \lambda_j v_j\rangle = \lambda_j v_i^T v_j
\end{aligned}$$

$(\lambda_i - \lambda_j)\underbrace{v_i^T v_j} = 0 \Rightarrow$ since $\lambda_i \neq \lambda_j$ we have that

$v_i^T v_j = 0 \Rightarrow v_i \perp v_j$

What we have shown so far...

- Optimality: $\nabla f(\theta^\star) = 0$ and $H(\theta^\star) \succ 0 \implies \theta^\star$ is a (local) minimum.

$H = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$

$v^\top H v = a v_1^2 + b v_2^2 > 0$

$$H(\theta) = \nabla^2 f(\theta) = \begin{bmatrix} \frac{\partial^2 f}{\partial \theta_1^2} & \cdots & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial \theta_d \partial \theta_1} & \cdots & \frac{\partial^2 f}{\partial \theta_d^2} \end{bmatrix} \qquad \text{(Hessian)}$$

$v^\top Q \Lambda Q^\top v$

$$H(\theta) \succ 0 \iff v^\top H(\theta) v > 0 \quad \forall\, v \neq 0 \implies f \text{ is } \textbf{strictly convex} \text{ at that point}$$

- **Eigenvalue test:** Since H is symmetric, we have that

$$H = Q \Lambda Q^\top, \quad Q^\top Q = I, \; \text{where } \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$$

- This means that H is positive definite if and only if all its eigenvalues are positive: $H \succ 0 \iff$ all eigenvalues $\lambda_i > 0$.

Eigenvalues and Curvature: PD, PSD, and Indefinite

For a symmetric matrix H, curvature along direction v is measured by

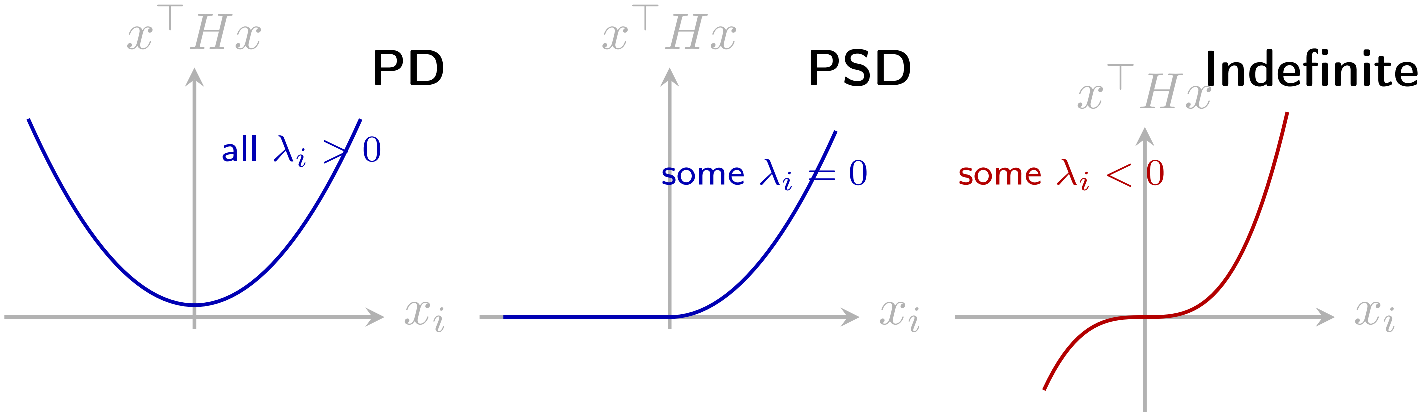
$$v^\top H v = \lambda \|v\|_2^2.$$

$\lambda_i > 0 \implies$ curves upward (bowl)

$\lambda_i = 0 \implies$ flat direction (ridge)

$\lambda_i < 0 \implies$ curves downward (saddle)

- curvature in every direction $\to$ unique minimum.
- flat direction $\to$ multiple minima or constraints.
- saddle or concave directions $\to$ not convex.



Computing Eigenvalues: The Characteristic Polynomial

Definition of Eigenvalues and Eigenvectors

Suppose we have $A \in \mathbb{R}^{n \times n}$. an eigenpair (λ, v) satisfy

$$Av = \lambda v \implies \underbrace{(A - \lambda I)\, v = 0}_{} \quad (1)$$

$$\mathrm{Null}(A - \lambda I) = \{ v \mid (A - \lambda I) v = 0 \}$$

A nontrivial soln $v \neq 0$ exists for (1) only when $A - \lambda I$ is not invertible for some λ

i.e. $\underbrace{\det(A - \lambda I)}_{} = 0$

$$= P_A(\lambda) = \chi_A(\lambda)$$

Characteristic polynomial

Aside

$Bw = z$

then a soln w exists if B is invertible $\iff \det(B) \neq 0$

Why $\det(A - \lambda I) = 0$ Defines Eigenvalues

Eigenvalue equation

$$Av = \lambda v \Rightarrow \underbrace{(A - \lambda I)v = 0}_{\text{Homogeneous linear system}}$$

For any Homogeneous linear system $Mw = 0$ has solns.

$$\begin{cases} w = 0 \Rightarrow \text{when } M \text{ is invertible this is the unique soln} \\ w \neq 0 \Rightarrow \text{when } M \text{ is singular (aka not invertible) aka } \det(M) = 0 \end{cases}$$

Fact

λ is an eigenvalue of $A \iff \det(A - \lambda I) = 0.$

Why $\det(A - \lambda I) = 0$ Defines the Eigenvalues

Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$p_A(\lambda) = \det(A - \lambda I) = \det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = \det\left(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}\right)$$

$$\det(A-\lambda I) = (2-\lambda)^2 - 1 = 0 \quad = \lambda^2 - 4\lambda + 3 \quad = (\lambda-1)(\lambda-3)$$

$$= (\lambda - \xi_1)(\lambda - \xi_2) \qquad \Rightarrow \lambda = 1, 3$$

$$Av = \lambda v$$

Takeaway

Eigenvalues are the roots of the polynomial $\det(A - \lambda I) = 0$.

3×3 Example

Example

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}. \qquad p_A(\lambda) = \det(A - \lambda I) = \det \begin{array}{c} \begin{array}{ccc} & b_1 & b_2 \end{array} \\ \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 1 & 2-\lambda \end{bmatrix} \end{array}.$$

3×3 Example: Eigendecomposition

Eigenvalues in Higher Dimensions: The Characteristic Polynomial

Characteristic polynomial

$$p_A(\lambda) = (-1)^n \lambda^n + \underbrace{c_{n-1}}\lambda^{n-1} + \cdots + \underbrace{c_1}\lambda + \underbrace{c_0}$$

depend on entries of A

- roots of $p_A(\lambda)$ are eigenvalues
- If $A = A^T$, all λ's are Real

Calculus & Linear Algebra Summary

$f = \|X\theta - y\|^2$

- **Optimality conditions:** $\nabla f(\theta^\star) = 0$ and $H(\theta^\star) \succ 0 \implies \theta^\star$ is a (local) minimum.
- **Symmetric matrices can be diagonalized:** $H = Q\Lambda Q^\top$ for some orthogonal matrix Q and diagonal matrix Λ with eigenvalues $\lambda_1, \ldots, \lambda_d$.
- **Positive definiteness:** $H \succ 0 \iff$ all eigenvalues $\lambda_i > 0$.
- **Computing eigenvalues:** Eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I) = 0$.

$Av = \lambda v$

$(x_i, y_i)_{i=1}^n$

Solving Least Squares (Coordinate Form)

For one data point: $f_\theta(x_i) = \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_d x_{id} = \theta^\top x$.

$$\text{minimize } \mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^n (y_i - \theta_1 x_{i1} - \theta_2 x_{i2} - \cdots - \theta_d x_{id})^2 \cancel{= \frac{1}{n}\sum_{i=1}^n \sum_{j=1}^n x_{ij} x}$$

Taking partial derivatives:

$$\frac{\partial \mathcal{L}(\theta)}{\partial \theta_j} = -\frac{2}{n}\sum_{i=1}^n x_{ij}\left(y_i - \sum_{k=1}^d \theta_k x_{ik}\right) = 0$$

Normal eqns:

$$\sum_{i=1}^n x_{ij} y_i = \sum_{i=1}^n x_{ij} \sum_{k=1}^d \theta_k x_{ik}$$

Solving Least Squares (Coordinate Form)

Linear form

$\nabla_x(b^T x) = b$

$\nabla_x(x^T b) = b^T$

Quad Form

$\nabla_x [x^T A x]$

$= (A^T + A)x$

if $A = A^T$,

$\nabla_x(\cdot) = 2Ax$

$$\mathcal{L}(\theta) = \frac{1}{n}\|X\theta - y\|^2 = \frac{1}{n}(X\theta - y)^T(X\theta - y)$$

$$= \frac{1}{n}(\theta^T X^T X\theta - y^T X\theta - \theta^T X^T y + y^T y)$$

$$\nabla\mathcal{L}(\theta) = -\frac{2}{n}X^T(y - X\theta)$$

$$= 0 = X^T y - X^T X\theta \implies \underbrace{X^T y}_{d} = \underbrace{X^T X}_{A}\theta$$

$$A\theta = d$$

If $X^T X = A$ is invertible the

$$\theta = \underbrace{(X^T X)^{-1}}_{\text{projection}} X^T y \qquad \nabla^2\mathcal{L}(\theta) \propto X^T X$$

Matrix Derivatives of Quadratic Forms

Function: $f(x) = x^\top Ax,$ where $A \in \mathbb{R}^{n\times n}$.
Step 1. Expand in coordinates:

$$f(x) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j.$$

Then the partial derivative with respect to x_k is

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^{n} a_{kj}x_j + \sum_{i=1}^{n} a_{ik}x_i.$$

Combine in vector form:

$$\nabla f = (A + A^\top)x.$$

If A is symmetric, $A = A^\top$, so:

$$\boxed{\nabla(x^\top Ax) = 2Ax.}$$

Solving Least Squares (Vector Form)

Recall the objective:

$$\mathcal{L}(\theta) = \frac{1}{n}\|y - X\theta\|_2^2 = \frac{1}{n}(y - X\theta)^\top (y - X\theta).$$

Solving Least Squares (Vector Form)

Recall the objective:

$$\mathcal{L}(\theta) = \frac{1}{n}\|y - X\theta\|_2^2 = \frac{1}{n}(y - X\theta)^\top (y - X\theta).$$

Coordinate vs. Vector Gradient (2D Example)

Consider two features: $x_i = (x_{i1}, x_{i2})$ and parameters $\theta = (\theta_1, \theta_2)$.

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2)^2 = \frac{1}{n}\|y - X\theta\|_2^2.$$

Coordinate form:

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -\frac{2}{n}\sum_{i=1}^{n} x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2),$$

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = -\frac{2}{n}\sum_{i=1}^{n} x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2).$$

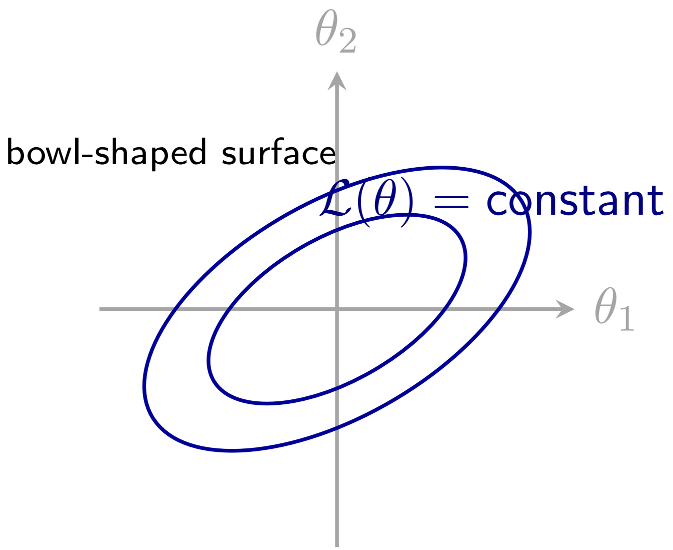
Vector form:

$$\begin{aligned}\nabla_\theta \mathcal{L}(\theta) &= -\frac{2}{n}X^\top(y - X\theta)\\ &= -\frac{2}{n}\begin{bmatrix}\sum_i x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2)\\ \sum_i x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2)\end{bmatrix}.\end{aligned}$$

Convexity of the Least Squares Objective

$$\mathcal{L}(\theta) = \frac{1}{n}\|y - X\theta\|_2^2 = \frac{1}{n}(y - X\theta)^\top (y - X\theta).$$

Convexity Intuition (2D Example)



Takeaway: Least squares is convex because its Hessian $\frac{2}{n}X^\top X$ is positive definite when X has full column rank.

What happens if some directions have zero curvature?

Back to the optimization problem...

What does it mean to be a solution?

Column Interpretation

$$X = \begin{bmatrix} | & \cdots & | \\ x_1 & \cdots & x_d \\ | & \cdots & | \end{bmatrix}, \quad x_i \in \mathbb{R}^n$$

Row Interpretation

$$X = \begin{bmatrix} — & \tilde{x}_1^\top & — \\ \vdots & \vdots & \vdots \\ — & \tilde{x}_n^\top & — \end{bmatrix}, \quad \tilde{x}_i \in \mathbb{R}^d$$

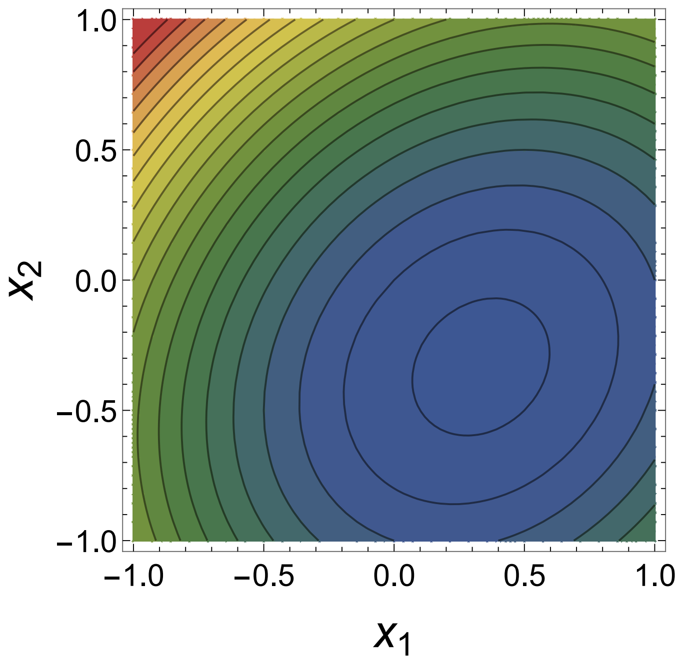
3×2 Example

Example

$$X = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad n = 3,\ d = 2$$

Example Continued

$$\min_{\theta_1,\theta_2}\{(2\theta_1-1)^2+(-x_1+\theta_2)^2+(2\theta_2+1)^2\}$$



Least Squares Solution Properties & Facts

Assumption [A1]:

The columns of X are linearly independent; i.e., $\sum_{i=1}^{n} c_i x_i = 0 \iff c_i = 0 \quad \forall i = 1, \ldots, n$

Why "Pseudo-Inverse"?

For a square invertible matrix A, the inverse A^{-1} satisfies $A^{-1}A = I$.

Why "Pseudo-Inverse"?

The Pseudo-Inverse is a Projection Matrix

The Pseudo-Inverse is a Projection Matrix

The matrix $P = X(X^\top X)^{-1}X^\top$ is the orthogonal projection onto the column space of X.

Properties:

- $P^2 = P$ (idempotent): Projecting twice = projecting once
- $P^\top = P$ (symmetric): Orthogonal projection
- Py is the closest point in $\text{col}(X)$ to y
- $(I - P)$ projects onto the orthogonal complement (residuals in regression)

In linear regression $\hat{\mathbf{y}} = X\hat{\theta} = X(X^\top X)^{-1}X^\top y = Py$ is the projection of y onto the column space.

Direct Verification that the Solution is a Minimum

Example Problem: Solution is a Minimum

Show that for any $\theta \neq \hat{\theta} = X^{\dagger} y$ we have

$$\|X\hat{\theta} - y\|_2^2 < \|X\theta - y\|_2^2$$

Direct Verification that the Solution is a Minimum

Projection

- The least squares solution is $\hat{\theta} = (X^\top X)^{-1} X^\top y$ and the prediction is $\hat{y} = X\hat{\theta}$
- $P = X(X^\top X)^{-1} X^\top$ is a projection matrix: it projects onto the subspace formed by the columns of X
  - P is a projection matrix if $P^2 = P$
- Orthogonal decomposition:

$$y = y_{\mathcal{R}(X)} + y_{\mathcal{R}(X)^\perp} \quad \text{where } \; y_{\mathcal{R}(X)} = X\hat{\theta} \text{ and } y_{\mathcal{R}(X)^\perp} = r = X\hat{\theta} - y$$

Example

Example: Over-determined System

Consider

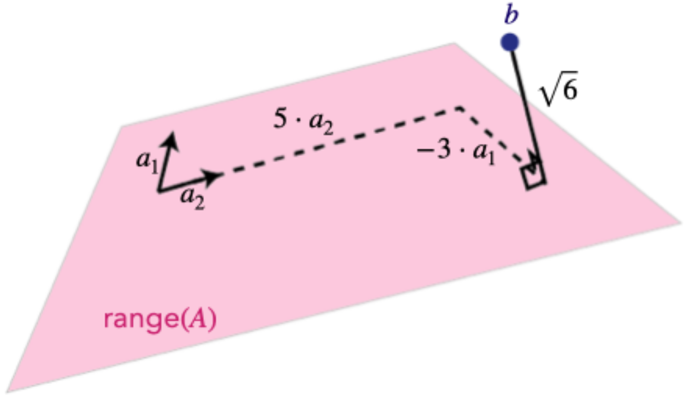
$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

Find the least squares approximate solution to $Ax = b$.

Example Continued

- The solution minimizes the distance from $A\hat{x}$ to b:

$$\|y - X\hat{\theta}\|_2^2 = \left\| \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\|_2^2$$



What about under-determined systems?

Minimum Norm Solution

Proof of Minimal Norm Property.

Null Space Connection

Pseudo-Inverse and Projection Formulas

Next time

Today we covered:

- Convexity, Hessians, and curvature
- Linear independence, basis, and dimension
- Why symmetric matrices can be diagonalized
- Checking convexity with the eigenvalue test
- Solving least squares by setting the gradient to zero

Next time we will cover:

- Under-determined systems
- (next week) regularization and ridge regression
- (next week) bias-variance tradeoff, and generalization error